

Below is the entirety of the output after running the MATLAB code to determine ideal placement of the mass blocks to reflect Cal Poly Pomona's 3U CubeSat Cerberus' Moment of Inertia (MOI) tensor.

=== CONFIG ===
Axis convention: X = width, Y = stacking (long), Z = depth
SW Mass: 2188.08 g | Infill Ratio: 0.3804 | Static Mass: 832.38 g
Allowed Masses (g): [0, 15, 27]

Starting Optimization (24 Trials)...
Initializing parallel pool...
Starting parallel pool (parpool) using the 'Processes' profile ...
Connected to parallel pool with 4 workers.
Optimization Progress: 100.0%
Selected trial 2: 157 blocks, cost 9.1885e+00 (within tolerances)

Phase 2: Snapping to exact allowed inventory blocks...
Phase 3: Pruning unnecessary blocks...
Pruning removed 0 additional blocks.
Phase 4: Reducing oversized blocks...
Downsizing reduced mass on 1 blocks.
Final result: 157 blocks, total tuning mass 4227.0 g

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3U CUBESAT INERTIA OPTIMIZATION REPORT

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Static Hardware Mass: 832 g
Active Tuning Blocks: 157

Metric	Target	Actual	Diff	Error (%)	Status
Mass (g)	INFO	5059.3800	-	-	N/A
COM X (cm)	0.0970	-0.0010	9.8020e-02	-	INFO
COM Y (cm)	-0.7110	-0.5667	1.4430e-01	-	INFO
COM Z (cm)	-0.2830	0.0157	2.9869e-01	-	INFO
I_xx (g*cm^2)	294229.6900	291602.6240	2.6271e+03	0.893%	PASS
I_yy (g*cm^2)	94578.5700	93650.2120	9.2836e+02	0.982%	PASS
I_zz (g*cm^2)	284732.3400	282017.1237	2.7152e+03	0.954%	PASS
I_xy (g*cm^2)	2840.5600	3494.4898	6.5393e+02	0.222%	PASS
I_xz (g*cm^2)	87.8600	141.9222	5.4062e+01	0.018%	PASS
I_yz (g*cm^2)	3663.8600	2834.9096	8.2895e+02	0.282%	PASS

--- ASSEMBLED INERTIA TENSOR (g*cm^2) ---
X [291602.62 3494.49 141.92]
Y [3494.49 93650.21 2834.91]
Z [141.92 2834.91 282017.12]

--- BLOCK PLACEMENT (Active Slots Only) ---

Mass (g)	X (cm)	Y (cm)	Z (cm)
27	-3.44	-13.42	-3.44
27	3.44	-13.42	-3.44
15	-3.44	-11.66	-3.44
27	1.72	-11.66	-3.44

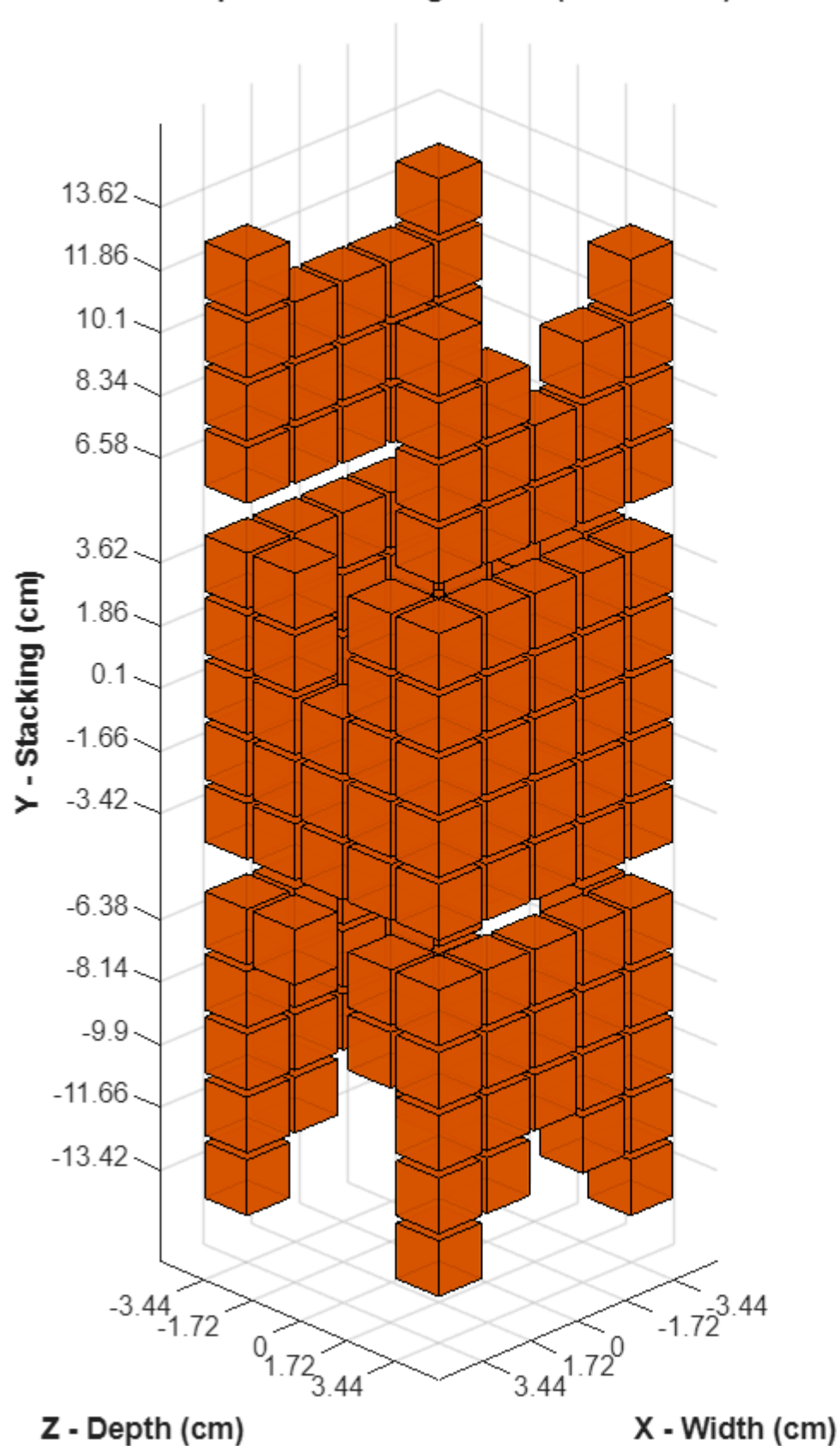
27	3.44	-11.66	-3.44	
27	-3.44	-9.90	-3.44	
27	-1.72	-9.90	-3.44	
27	0.00	-9.90	-3.44	
27	1.72	-9.90	-3.44	
27	3.44	-9.90	-3.44	
27	-3.44	-8.14	-3.44	
27	-1.72	-8.14	-3.44	
27	0.00	-8.14	-3.44	
27	1.72	-8.14	-3.44	
27	3.44	-8.14	-3.44	
27	-3.44	-6.38	-3.44	
27	-1.72	-6.38	-3.44	
27	0.00	-6.38	-3.44	
27	1.72	-6.38	-3.44	
27	3.44	-6.38	-3.44	
27	-3.44	-3.42	-3.44	
27	-1.72	-3.42	-3.44	
27	0.00	-3.42	-3.44	
27	1.72	-3.42	-3.44	
27	3.44	-3.42	-3.44	
27	-3.44	-1.66	-3.44	
27	-1.72	-1.66	-3.44	
27	0.00	-1.66	-3.44	
27	1.72	-1.66	-3.44	
27	3.44	-1.66	-3.44	
27	-3.44	0.10	-3.44	
27	-1.72	0.10	-3.44	
27	0.00	0.10	-3.44	
27	1.72	0.10	-3.44	
27	3.44	0.10	-3.44	
27	-3.44	1.86	-3.44	
27	-1.72	1.86	-3.44	
27	0.00	1.86	-3.44	
27	1.72	1.86	-3.44	
27	3.44	1.86	-3.44	
27	-3.44	3.62	-3.44	
27	-1.72	3.62	-3.44	
27	0.00	3.62	-3.44	
27	1.72	3.62	-3.44	
27	3.44	3.62	-3.44	
27	-3.44	6.58	-3.44	
27	-1.72	6.58	-3.44	
27	0.00	6.58	-3.44	
27	1.72	6.58	-3.44	
27	3.44	6.58	-3.44	
27	-3.44	8.34	-3.44	
27	-1.72	8.34	-3.44	
27	0.00	8.34	-3.44	
27	1.72	8.34	-3.44	
27	3.44	8.34	-3.44	
27	-3.44	10.10	-3.44	
27	-1.72	10.10	-3.44	
27	0.00	10.10	-3.44	
27	1.72	10.10	-3.44	
27	3.44	10.10	-3.44	
27	-3.44	11.86	-3.44	
27	3.44	11.86	-3.44	
27	-3.44	-6.38	-1.72	

27	3.44	-6.38	-1.72	
27	-3.44	-3.42	-1.72	
27	3.44	-3.42	-1.72	
27	-3.44	-1.66	-1.72	
27	3.44	-1.66	-1.72	
27	-3.44	0.10	-1.72	
27	3.44	0.10	-1.72	
27	-3.44	1.86	-1.72	
27	3.44	1.86	-1.72	
27	-3.44	3.62	-1.72	
27	3.44	3.62	-1.72	
27	-3.44	6.58	-1.72	
27	3.44	-3.42	0.00	
27	-3.44	-1.66	0.00	
27	3.44	-1.66	0.00	
27	-3.44	0.10	0.00	
27	3.44	0.10	0.00	
27	-3.44	1.86	0.00	
27	3.44	-8.14	1.72	
27	-3.44	-6.38	1.72	
27	3.44	-6.38	1.72	
27	-3.44	-3.42	1.72	
27	3.44	-3.42	1.72	
27	-3.44	-1.66	1.72	
27	3.44	-1.66	1.72	
27	-3.44	0.10	1.72	
27	3.44	0.10	1.72	
27	-3.44	1.86	1.72	
27	3.44	1.86	1.72	
27	-3.44	3.62	1.72	
27	3.44	3.62	1.72	
27	-3.44	6.58	1.72	
27	-3.44	-13.42	3.44	
27	3.44	-13.42	3.44	
27	-3.44	-11.66	3.44	
27	-1.72	-11.66	3.44	
27	1.72	-11.66	3.44	
27	3.44	-11.66	3.44	
27	-3.44	-9.90	3.44	
27	-1.72	-9.90	3.44	
27	0.00	-9.90	3.44	
27	1.72	-9.90	3.44	
27	3.44	-9.90	3.44	
27	-3.44	-8.14	3.44	
27	-1.72	-8.14	3.44	
27	0.00	-8.14	3.44	
27	1.72	-8.14	3.44	
27	3.44	-8.14	3.44	
27	-3.44	-6.38	3.44	
27	-1.72	-6.38	3.44	
27	0.00	-6.38	3.44	
27	1.72	-6.38	3.44	
27	3.44	-6.38	3.44	
27	-3.44	-3.42	3.44	
27	-1.72	-3.42	3.44	
27	0.00	-3.42	3.44	
27	1.72	-3.42	3.44	
27	3.44	-3.42	3.44	
27	-3.44	-1.66	3.44	

27	-1.72	-1.66	3.44	
27	0.00	-1.66	3.44	
27	1.72	-1.66	3.44	
27	3.44	-1.66	3.44	
27	-3.44	0.10	3.44	
27	-1.72	0.10	3.44	
27	0.00	0.10	3.44	
27	1.72	0.10	3.44	
27	3.44	0.10	3.44	
27	-3.44	1.86	3.44	
27	-1.72	1.86	3.44	
27	0.00	1.86	3.44	
27	1.72	1.86	3.44	
27	3.44	1.86	3.44	
27	-3.44	3.62	3.44	
27	-1.72	3.62	3.44	
27	0.00	3.62	3.44	
27	1.72	3.62	3.44	
27	3.44	3.62	3.44	
27	-3.44	6.58	3.44	
27	-1.72	6.58	3.44	
27	0.00	6.58	3.44	
27	1.72	6.58	3.44	
27	3.44	6.58	3.44	
27	-3.44	8.34	3.44	
27	-1.72	8.34	3.44	
27	0.00	8.34	3.44	
27	1.72	8.34	3.44	
27	3.44	8.34	3.44	
27	-3.44	10.10	3.44	
27	-1.72	10.10	3.44	
27	1.72	10.10	3.44	
27	3.44	10.10	3.44	
27	-3.44	11.86	3.44	
27	3.44	11.86	3.44	

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3U Optimized Configuration (157 Blocks)



Block Mass	
	15 g
	27 g
	Target COM
	Actual COM